

Approximating Offensive Box Plus-Minus

Using regression and box-score stats to estimate a rigorous statistic

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Introduction

In previous research projects, I encountered a situation in which there was an advanced basketball metric that I needed, but it wasn't available. In my [Coach Hurley ASU retrospective](#), I needed Kenpom adjusted ratings from the 2010s, but I didn't have access. I approximated them using Kenpom ratings from 2024¹ and *Sports-Reference* metrics, using the actual ratings to derive coefficients from the readily available *Sports-Reference* metrics.

In my March Madness bracket, I needed to calculate two advanced metrics (*SRS* and *SOS*) from games before the tournament began. Those figures were not available, so I was forced to use the [Pro-Football-Reference](#) article about calculating the two metrics and [replicate it within my own dataset](#). They laid out the process, but I had to perform linear algebra on matrices with more than 350 rows over fifteen years. It was very time-consuming.

Advanced statistics are valuable, but obtaining them can be expensive. It would be much easier to approximate these metrics that are difficult to calculate. My goal was to use linear regression to estimate *Basketball-Reference's* **Offensive Box Plus Minus** *OBPM* statistic using easily available box score statistics and use two different test sets to evaluate performance.

Defining OBPM

OBPM - "A box score estimate of the offensive points per 100 possessions a player contributed above a league-average player, translated to an average team" [1].

The [article](#) that details the computation process is about ten pages long, so this statistic is a prime candidate for approximation. Manually calculating *SRS* was a lengthy and arduous endeavor, which led to my motivation for estimating *OBPM*.

That article also contains a brief guide to interpret *BPM* [2], as there exists a *DBPM* as well:

- +10.0 is an all-time season (think peak Jordan or LeBron)
- +8.0 is an MVP season (think peak Dirk or peak Shaq)
- +6.0 is an all-NBA season

¹I had them locally saved for a March Madness project back in the day.

- +4.0 is in all-star consideration
- +2.0 is a good starter
- +0.0 is a decent starter or solid 6th man
- -2.0 is a bench player (this is also defined as "replacement level")
- Below -2.0 are many end-of-bench players

Modeling Setup

The data is from the 2022-23 NBA regular season; I reused a data set from a college stats project. In compliance with *Basketball-Reference's* [statistical minimums](#), I only included players who played at least 1500 minutes in 2022-23. Given that the goal was to use easily accessible box score stats, the pool of predictors was narrowed to the following:

- Minutes per game
- Free Throw Percentage ($FT\%$)
- Two-Point Field Goal Percentage ($2P\%$)
- Three-Point Field Goal Percentage ($3P\%$)
- Effective Field Goal Percentage ($eFG\%$)
- True Shooting Percentage ($TS\%$)
- Two-Point Field Goal Attempts per 36 minutes ($2PA/36$)
- Three-Point Field Goal Attempts per 36 minutes ($3PA/36$)
- Free Throw Attempts per 36 minutes ($FTA/36$)
- Points per 36 minutes ($PTS/36$)
- Assists per 36 minutes ($AST/36$)
- Rebounds per 36 minutes ($REB/36$)
- Blocks per 36 minutes ($BLK/36$)
- Steals per 36 minutes ($STL/36$)
- Turnovers per 36 minutes ($TOV/36$)
- Position (as a numeric factor, 1 = PG, 2 = SG, ...)

I traded totals and game averages for minute-adjusted metrics, since *OBPM* is not a cumulative stat, but rather a measure per 100 possessions.

Results

In 2022-23, 190 NBA players reached the 1500-minute threshold. That is a reasonable sample size, so I used an 80-20 train-test split before training the model.

I used forward and backward robust regression to find the best subset of variables. Obviously, there was rampant multicollinearity within the pool of variables detailed above that needed to be addressed. The shooting percentages, attempts, and even points have several positive relationships. After addressing multicollinearity and removing insignificant predictors, I had two candidates for the final model:

$$\text{Backward: } OBPM \sim MPG + TS\% + FTA/36 + AST/36 + REB/36 \\ + 2PA/36 + 3PA/36 + STL/36 + TOV/36$$

$$\text{Forward: } OBPM \sim PTS/36 + TS\% + AST/36 + TOV/36 + REB/36 \\ + MPG + 3P\%$$

A few things to note:

- Predictors that are in both models: MPG , $TS\%$, $AST/36$, $TOV/36$, $REB/36$
- The backward model doesn't include points; it instead includes that information with $2PA$, $3PA$, FTA , and $TS\%$, meaning it indirectly contains points.
- The backward model contains steals? I was confused at first, but steals can certainly help offenses. Steals generate easy baskets and fast break scoring chances at the very least. $STL/36$ had a positive coefficient (0.444), which reinforces this insight.
- The forward model contains both $TS\%$ and $3P\%$, but three-pointers are not directly included within the true shooting formula, and their VIF values reflect their lack of multicollinearity (1.417 vs. 1.634).

Here's a table of the models' metrics on the training data.

Model	p	RSS	Adj. R^2	AIC	Mallow's C_p	MSE
Backward	9	54.5	0.9371	297.3	50.1	0.3583
Forward	7	56.3	0.9359	298.3	52.2	0.3702

Table 1: Training set metrics

The backward model has a slight edge, but with the cost of two additional predictors. The gap in metrics is not extremely significant, as reflected by the sequential ANOVA table.

Model	Res. Df	RSS	Df	Sum of Squares	F	p -value
Null	151	920.12				
Forward	144	56.28	7	863.84	412.6736	$< 2.2e-16$
Backward	142	54.46	2	1.81	3.0339	0.0515
Full Model	132	39.47	10	14.99	5.0124	3.676e-06

Table 2: ANOVA table containing the null, backward, forward, and full regression model.

The backward model is very strong, but given how close the forward model is in terms of performance, it is the better pick. Let's see how the two models performed on the test set of 38 players.

Model	Test MSE	Test MAE	Test Adj. R^2
Backward	0.4871	0.5390	0.9006
Forward	0.4858	0.5460	0.9075

Table 3: How the two candidates performed on the test set.

To avoid unfortunate seed luck, I ran 10-fold cross validation on the two model candidates and returned the average of the test MSE , MAE and adjusted R^2 :

Model	Test MSE	Test MAE	Test Adj. R^2
Backward	0.4223	0.5067	0.8564
Forward	0.4274	0.5065	0.8740

Table 4: 10-Fold Cross Validation Results

Given the extremely comparable results on unfamiliar data and the lower complexity, I am partial to the forward model.

Predictor	Estimate	Std. Error	p -value	VIF
Intercept	-18.8651	0.9139	7.191e-45	
$PTS/36$	0.3449	0.0158	1.972e-47	3.2560
$TS\%$	17.2297	1.3527	2.309e-25	1.4175
$AST/36$	0.6185	0.0356	3.759e-37	2.5660
$TO/36$	-1.4173	0.1282	5.881e-21	3.8371
$REB/36$	0.2186	0.0254	1.181e-14	2.1310
MPG	0.0396	0.0158	0.0132	2.0737
$3P\%$	2.7981	0.6937	8.875e-5	1.6338

Table 5: Forward Model Coefficient Table

A New Season

For one last assessment, I imported the 2023-24 NBA dataset to test the model's performance on a different season.

MSE	MAE	Adj. R^2	Spearman's ρ	Kendall's τ
0.8431	0.7065	0.8446	0.7586 ¹	0.9028 ²

Table 6: How the Forward model performed on unseen data from 2023-24

The MSE and MAE both increased, but given that these are unfamiliar data on an entirely different season, that is still very solid. The plot (Figure 1) shows a strong adherence between the fitted and actual values of $OBPM$. The Spearman and Kendall

¹ p -value ≈ 0

² p -value ≈ 0

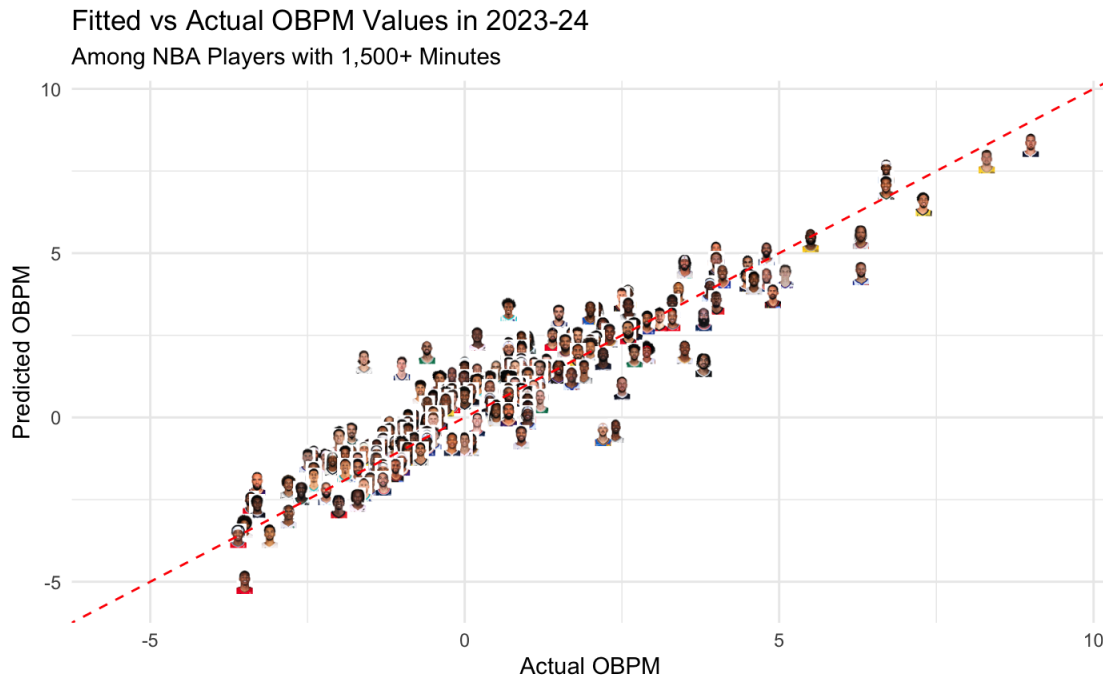


Figure 1: Fitted vs Actual OBPM in 2023-24

values also indicate that the relationship between the actual and fitted values has a statistically significant correlation, confirming the eye test in Figure 1.

Despite strong performance in a brand new season, I must acknowledge that the model has limitations in its prediction power, due to the era in which it was built.

Limitations by Era

In both the Backward and Forward models, there are variables dedicated to shooting efficiency ($TS\%$) and three-point shooting ($3PA/36$ in Backward, $3P\%$ in Forward). The three-point shot has surged from gimmick to gameplan since the 1980s, so the weight and significance of those variables in the 2020s will not accurately reflect the weight of the 1980s. Here's what happened when I applied the Forward model to the 1982-83 season:

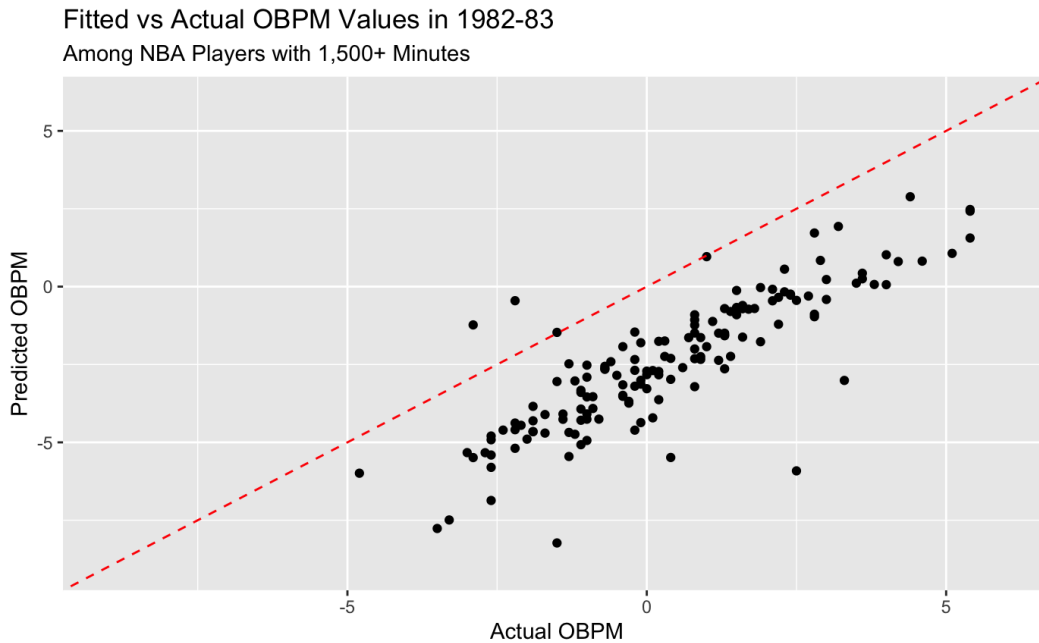


Figure 2: Applying the Forward model to the 1982-83 Season

This is a hilarious image; Figure 2 looks as if the predicted *OBPM* should be vertically adjusted by about two or three. It reflects the limitations of predicting *OBPM* placed by the era in which the model was trained. The NBA undergoes constant evolution, so a model of this kind will age poorly and won't be appropriate in a different era.

Conclusion & Future Ideas

I learned a few things from this exercise. First, it is possible to predict *OBPM*, a statistic that requires ten pages of documentation, with statistics found in a box score. As seen by the models' performance on unfamiliar data, there are genuine patterns within those box score statistics.

However, given the evolution undergone in NBA offenses and playstyles, a model that performs well in 2023 will not perform well in 1983, as seen in Figure 2. Given that reality, there are a few ideas that I would potentially explore in the future:

- Include season (either decade, era, etc.) as a factor in the regression.
- Include an interaction term between season and three-point shooting, even their undeniable relationship.
- Less fun answer: Make separate models for separate time-spans/decades.

References

- [1] “Glossary,” *Basketball-Reference*, [URL](#).
- [2] Myers, Daniel, “About Box Plus/Minus (BPM),” *Basketball-Reference*, Feb. 2020, [URL](#).
- [3] “Rate Statistic Requirements,” *Basketball-Reference*, [URL](#).
- [4] Data aggregation and cleaning was done with Excel, Python, and R.
- [5] Statistical analysis and data visualization was done using R tidyverse and stats.